UTD Secure HSM Design

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Abstract. This document describes the security architecture for UT Dallas's submission to the 2026 MITRE Embedded Capture the Flag (eCTF) competition. The system implements a permissioned Hardware Security Module (HSM) for securely storing and transferring IC fabrication data. The design uses AES-256-GCM authenticated encryption via the MSPM0L2228 AESADV hardware peripheral for data at rest and in transit, HMAC-SHA256 (wolfcrypt) for mutual authentication and permission verification, hardware-TRNG-derived nonces and challenges, and a suite of countermeasures against fault injection and side-channel analysis.

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1 System Overview

1.1 Competition Context

The 2026 eCTF scenario has teams design a Hardware Security Module (HSM) for ChipCorp's confidential IC fabrication workflow. Each deployed set of HSMs consists of three roles:

- **Engineer HSM** — Holds design files; has full read/write/receive permissions for the design permission group. PIN unknown to attacker.
- **Technician HSM** — Holds firmware update and calibration data; no permission to read or receive design files. PIN known to attacker.
- **Photolithography Machine HSM** — Receives files from both other HSMs; attacker has remote-only transfer-interface access.

Each HSM is built from the same global secrets file but with distinct permissions and PIN, yielding unique `PERMISSION_MAC` and `PIN_HMAC` values per device. `PIN_KEY` is generated independently per HSM inside `secrets_to_c_header.py` so that a CPA recovery of one device's PIN key does not affect any other device.

2 Security Requirements Compliance

SR	Requirement	Implementation
SR1	No file action without a validly provisioned HSM with the correct permission for that group	HMAC challenge-response; `PERMISSION_MAC`; `validate_permission()` double-pass
SR2	No PIN-protected action completable without prior knowledge of the PIN; PIN must not be exposed	`SECURE_PIN_CHECK` at every management-interface handler entry; PIN never stored in flash
SR3	HSM must not receive any file not generated by a valid HSM with write permission for that group	4-round mutual HMAC handshake; GCM tag over Transfer AAD binding challenges, slot, UUID, group, and name
SR4	File integrity must be cryptographically protected	AES-256-GCM tag; AAD binds slot, UUID, group, and name at rest and in transit
SR5	Files must be confidential at rest and in transit	`STORAGE_KEY` (at rest); `TRANSFER_KEY` (wire); independent 256-bit keys

3 Threat Model

The attacker has full physical access, a ChipWhisperer power-analysis platform, and the complete source code for the Technician and Engineer HSMs. The attacker knows the Technician PIN (`1a2b3c`) but not the Engineer PIN. The Photolithography Machine is accessible only via its UART1 transfer interface; no physical access or management interface is available to the attacker.

3.1 Attack Goals

- **Read Design** — Read `design_file1` from Engineer HSM without the Engineer PIN.
- **Steal Design** — Cause Engineer HSM to transfer `design_file1` to a device without Design receive permission.
- **Read Update** — Read `update_file` from Technician HSM without read permission on the Update group.
- **Compromise Machine** — Cause Photolithography HSM to accept a corrupted update file via the transfer interface.
- **Backdoored Design** — Cause Photolithography HSM to accept a specially crafted design file via the transfer interface.

4 Cryptographic Architecture

4.1 Key Split

`gen_secrets.py` generates five global 256-bit keys using Python's `secrets.token_bytes(32)` (CSPRNG backed by `/dev/urandom`). `CHALLENGE_TWEAK` is a 12-byte deployment-wide constant from the same source. `PIN_KEY` is generated independently per device inside `secrets_to_c_header.py` using `os.urandom(32)`.

Key	Size	Scope	Purpose
`STORAGE_KEY`	256 bit	Global	AES-256-GCM for files at rest
`TRANSFER_KEY`	256 bit	Global	AES-256-GCM for files in transit
`TRANSFER_AUTH_KEY`	256 bit	Global	HMAC-SHA256 for protocol challenge-response only
`PERM_MAC_KEY`	256 bit	Global	HMAC-SHA256 for `PERMISSION_MAC` only (build-time; never enters runtime hardware)
`CHALLENGE_TWEAK`	12 byte	Global	XOR mask applied to every handshake nonce before HMAC; breaks chosen-input CPA on `TRANSFER_AUTH_KEY`
`PIN_KEY`	256 bit	Per-device	HMAC-SHA256 for PIN verification; independent per HSM

Rationale. Separating storage and transfer keys means a CPA campaign against AES during READ reveals only `STORAGE_KEY`, giving no information about `TRANSFER_KEY` or any HMAC key. `PERM_MAC_KEY` is architecturally isolated from `TRANSFER_AUTH_KEY`: an attacker performing CPA on handshake traces can at most recover `TRANSFER_AUTH_KEY`, which does not let them forge a `PERMISSION_MAC` (signed with the build-time-only `PERM_MAC_KEY`). `CHALLENGE_TWEAK` XOR'd into every nonce before the HMAC call removes the attacker's ability to choose the actual HMAC first-block input even when they control the wire nonce, eliminating the chosen-input advantage required for standard CPA. `PIN_KEY` is per-device so recovering one device's PIN key reveals nothing about any other device.

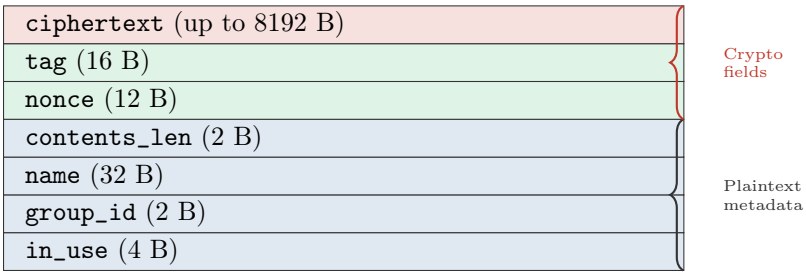
4.2 Algorithms

- **Symmetric encryption + integrity:** AES-256-GCM via the MSPM0L2228 `AESADV` hardware peripheral in `DL_AESADV_MODE_GCM_AUTONOMOUS`. The hardware derives GHASH subkey H and computes Y_0-encrypted internally; no separate ECB pass is needed. The peripheral is reset after every call to clear key registers. CPU-polling mode (DMA disabled); data fed in 16-byte aligned blocks.
- **MAC / authentication:** HMAC-SHA256 via wolfcrypt (`wc_HmacSetKey`, `wc_HmacUpdate`, `wc_HmacFinal`). The MSPM0L2228 has no hardware HMAC accelerator.
- **Nonce generation:** Hardware TRNG (`DL_TRNG`) producing 12-byte nonces per operation. Nonce reuse probability: $\approx 2^{-96}$.
- **Domain separation:** All HMAC calls include a mandatory, null-terminated domain label appended to the data (e.g., `"pin"`, `"sender"`, `"interrogate_req"`). No domain-free HMAC variant exists.

5 Data Structures

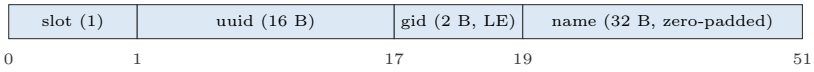
5.1 Stored File (`file_t`)

Files at rest in flash have the following layout. Only `ciphertext` is encrypted; the header fields are integrity-protected via the GCM AAD and tag.



5.2 AAD Formats

Storage AAD (51 bytes) binds ciphertext to its slot, UUID, group, and filename:



Transfer AAD (75 bytes) additionally binds both challenges, slot, and UUID:

recv_chal (12)	send_chal (12)	sl	uuid (16)	gid (2)	name (32 B)

0 12 24 25 41 43 75

5.3 Wire Permission Format

Permissions are serialized as 5 bytes per entry: `group_id[2 LE] || read[1] || write[1] || receive[1]`. Boolean fields must be exactly `0x00` or `0x01` (enforced by `validate_bool()`).

6 Protocol Diagrams

6.1 PIN Verification

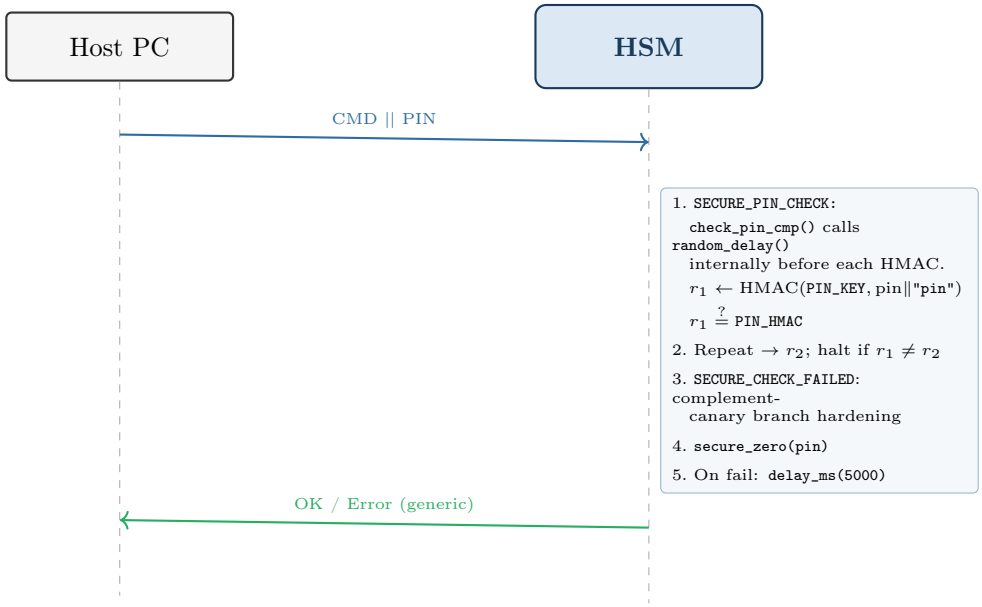


Figure 1: PIN verification flow. `random_delay()` fires inside each `check_pin_cmp()` call, so both passes have independent jitter. `SECURE_CHECK_FAILED` uses complement-canary hardening to resist a single branch-skip glitch after the double-pass.

6.2 RECEIVE Protocol (Mutual Authentication)

The RECEIVE command performs a 4-round mutual authentication handshake using HMAC-SHA256 and then transfers the file under AES-256-GCM with `TRANSFER_KEY`. The *Initiator* is the requesting HSM; the *Responder* holds the file. All HMAC calls on both sides apply `CHALLENGE_TWEAK` (XOR with nonce) before the hash to defeat chosen-input CPA.

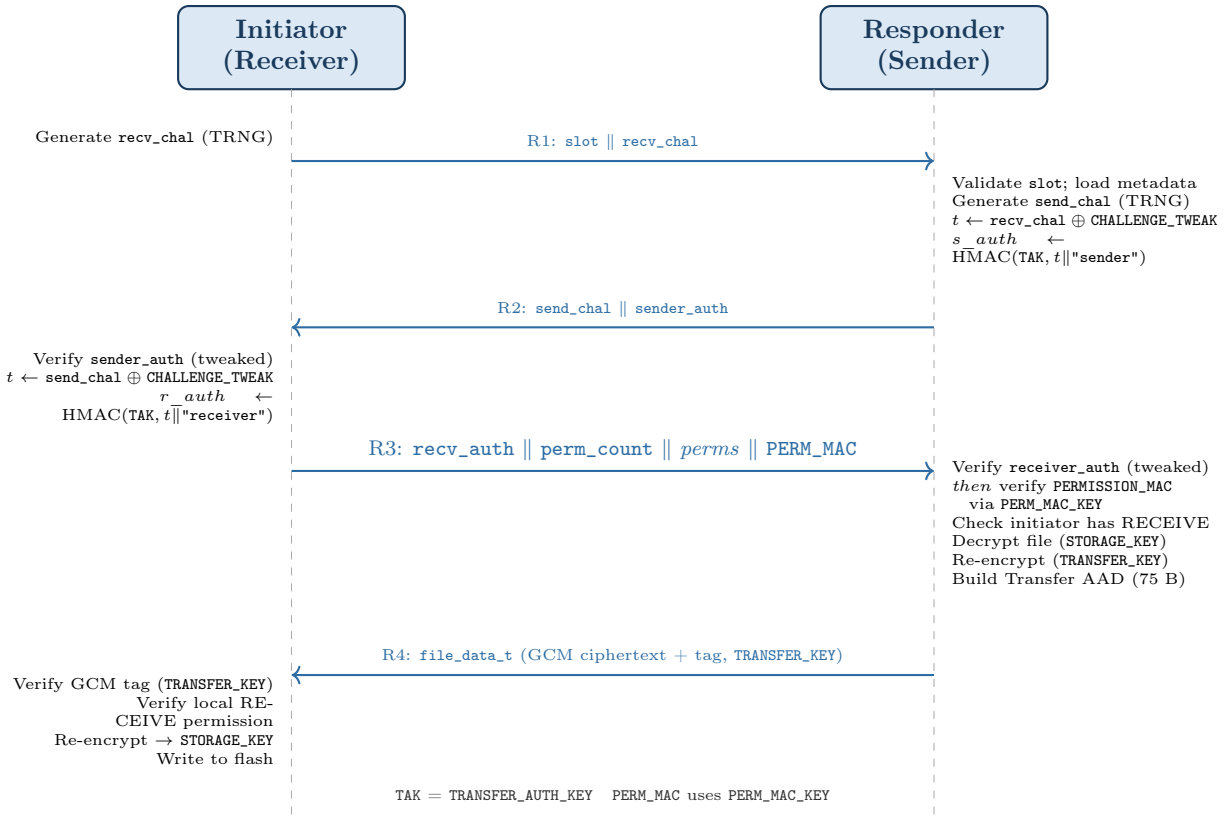


Figure 2: RECEIVE mutual authentication protocol. `TRANSFER_AUTH_KEY` is used only for challenge-response; `PERM_MAC_KEY` signs and verifies `PERMISSION_MAC`. `receiver_auth` is verified *before* `PERMISSION_MAC` so an unauthenticated sender cannot reach `PERM_MAC_KEY` operations with chosen permission bytes.

6.3 INTERROGATE Protocol

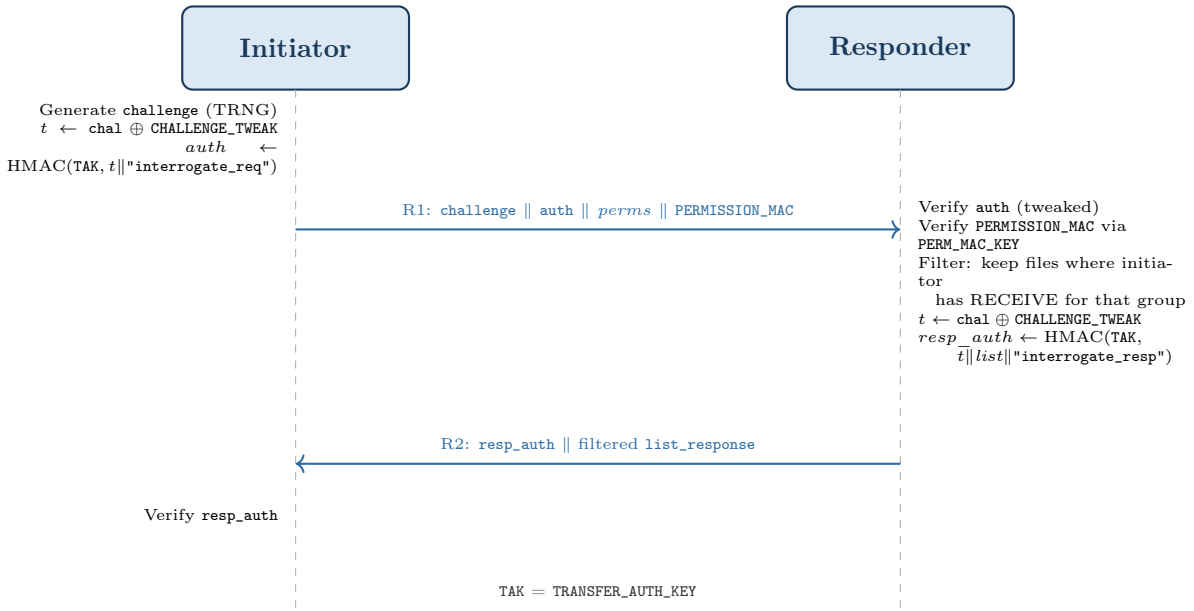


Figure 3: INTERROGATE protocol. The tweaked challenge binds the response, preventing a rogue responder from returning a cached or fabricated file list. `PERM_MAC_KEY` verifies permissions; `n_files` is bounds-checked before forwarding to the host.

7 Command Reference

All management-interface commands (LIST, READ, WRITE, RECEIVE, INTERROGATE) are served on UART0 and are PIN-protected. LISTEN is served on UART1 (transfer interface) and requires no PIN; it handles exactly one peer command per invocation. Messages arriving on the wrong interface are rejected with a generic error before any processing occurs.

7.1 READ Flow

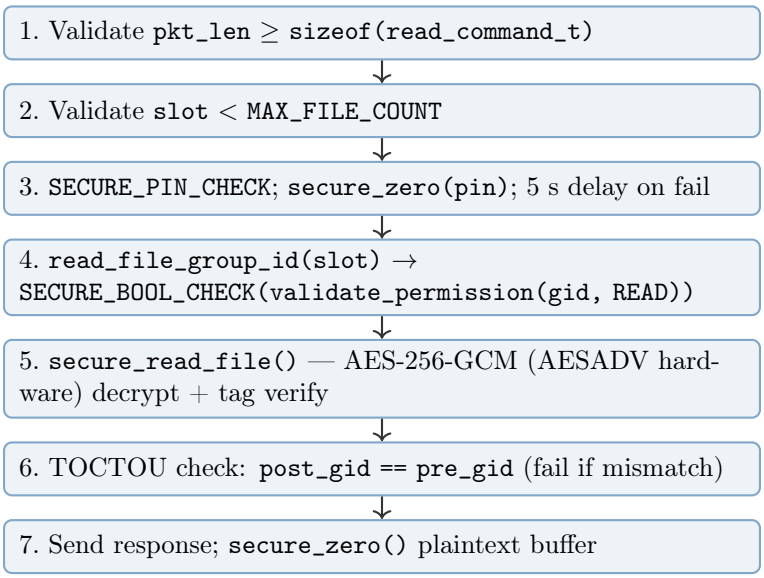


Figure 4: READ command flow. Permission is checked before decryption; a TOCTOU re-check ensures the group ID was not altered between reads. The WRITE flow is symmetric: PIN check → permission(WRITE) check → AES-256-GCM encrypt → store to flash.

8 Security Countermeasures

8.1 PIN Protection

- **HMAC-based storage.** The PIN is never stored in flash; only `PIN_HMAC` (see §4) is stored. Even if an attacker reads both from flash, offline brute-force requires $16^6 \approx 16$M HMAC evaluations without any on-device oracle.
- **Double-evaluation.** `SECURE_PIN_CHECK` calls `check_pin_cmp()` twice. If both results disagree, `security_halt()` fires immediately. A single fault injection cannot bypass both passes simultaneously.
- **Complement-canary branch hardening.** `SECURE_CHECK_FAILED` re-reads `ok1` and `ok2` from stack into separate `volatile` temporaries, computes their AND, and stores the bitwise complement as a canary. The branch passes only when the AND is true *and* the canary is its complement. A glitch corrupting one load or the AND cannot simultaneously corrupt the canary, so a single fault cannot silently accept a failed check.
- **Rate limiting.** A wrong-PIN result incurs a 5-second delay before the generic error is returned. Brute-force over the serial interface would take ≈ 2 years.
- **Immediate zeroing.** `secure_zero(pin)` is called at every check site before any other work proceeds.

8.2 Permission Enforcement

`validate_permission()` iterates `global_permissions[]` with a double-pass pattern: both passes must agree or the device halts. `SECURE_BOOL_CHECK` wraps every call site. On the transfer interface, the listening HSM re-verifies its own local permissions for the file's group after decryption (`PERM_RECEIVE` check in RECEIVE, filtered list in INTERROGATE). `receiver_auth` (HMAC with `TRANSFER_AUTH_KEY`) is verified *before* `PERMISSION_MAC` in the R3 block so that an unauthenticated sender cannot reach `PERM_MAC_KEY` operations with chosen permission bytes.

8.3 Side-Channel Attack Mitigations

Countermeasure	Details
Constant-time comparison	`secure_compare()` uses an XOR accumulator over the full buffer. No early exit. Used for all HMAC tag and GCM tag comparisons.
TRNG jitter (32 768 steps)	`random_delay()` reads *two* TRNG bytes: a coarse byte producing 0–≈4 ms jitter and a fine byte adding 0–≈1 ms, yielding $128 \times 256 = 32\,768$ distinct delay values. It is called inside `hmac_sha256()` before every `wc_HmacSetKey` and inside `aes_gcm_encrypt`/`decrypt` before every key load, so all HMAC and AES call sites are covered automatically.
Challenge tweak (chosen-input CPA)	`CHALLENGE_TWEAK` is XOR'd with every handshake and interrogate nonce before it enters `hmac_sha256()`. The attacker controls the wire nonce but cannot predict the actual HMAC first-block input, removing the chosen-input advantage required for CPA against `TRANSFER_AUTH_KEY`.

8.4 Memory Safety

- All packet sizes are validated against known struct sizes at entry.
- `slot` $<$ `MAX_FILE_COUNT` enforced before any flash access.
- `contents_len` $\leq$ `MAX_CONTENTS_SIZE` enforced at WRITE and RECEIVE.
- Name fields: printable ASCII only (`0x20`–`0x7E`), at least one character, null-terminated, bounded by `MAX_NAME_SIZE`.
- Permission boolean bytes in received proofs: validated to be exactly `0x00` or `0x01` by `validate_bool()`.
- `perm_count` $\leq$ `MAX_PERMS` enforced before any permission iteration.
- Cross-packet length check in RECEIVE: packet length must equal `FILE_DATA_HEADER_SIZE + contents_len`.
- `n_files` in INTERROGATE R2 is bounds-checked against the received `list_data_len` before forwarding to the host.
- `secure_zero()` called on all sensitive stack buffers before return.

8.5 Stack Canary

The firmware is compiled with `-fstack-protector-strong`. The canary (`__stack_chk_guard`) is seeded from the hardware TRNG on boot: the low byte is forced to `0x00` so any upward string-copy overwrite corrupts the canary before reaching the return address, and the upper 24 bits are random TRNG bytes to defeat non-string overwrites. `__stack_chk_fail()` delegates immediately to `security_halt()`.

9 Attack Mitigation Summary

Attack	Mitigation
Brute-force PIN via serial	5 s delay per failure; 2000 ms UART timeout per byte; 16^6 space takes ≈2 years online.
CPA against AES keys	AESADV hardware; `random_delay()` (0–≈5 ms, 32 768 steps) before every key load; peripheral reset after each call; four independent keys so one compromise reveals nothing about others.
CPA against HMAC keys	`random_delay()` inside every `hmac_sha256()` call; `CHALLENGE_TWEAK` XOR breaks chosen-input advantage on `TRANSFER_AUTH_KEY`; `PERM_MAC_KEY` never enters runtime hardware; `PIN_KEY` is per-device.
Offline PIN brute-force	16^6 HMAC evaluations required even with `PIN_KEY` and `PIN_HMAC` in hand; per-device key limits blast radius.
Single-glitch bypass (PIN, permission, GCM tag)	Double-pass evaluation on all three; `SECURE_CHECK_FAILED` complement-canary hardening on each branch; volatile double-check of GCM hardware return value and tag booleans (FI2/FI3); mismatch calls `security_halt()`.
Unauthenticated sender choosing permission bytes	`receiver_auth` verified *before* `PERMISSION_MAC` in R3; unauthenticated sender cannot reach `PERM_MAC_KEY`.
Replay of previous transfer	Fresh TRNG challenge per session; challenges bound into Transfer AAD.
Rogue device / permission forgery	Mutual HMAC challenge-response; `PERMISSION_MAC` = HMAC(`PERM_MAC_KEY`, ...) never recoverable via runtime CPA; GCM tag authenticates file content.
Metadata manipulation (group, slot, name)	All three fields bound into both storage AAD and transfer AAD; GCM tag fails on any alteration.
TOCTOU race on READ	`group_id` re-verified after decryption; mismatch returns error.
Stack smash	`-fstack-protector-strong`; TRNG-seeded canary with embedded null byte defeats string-copy overwrites.